ChatGPT

Overview

Livepeer is a decentralized platform for live and on-demand video computing. It powers an open video infrastructure ("open video") where anyone can upload, transcode, and stream video content using a globally distributed network of nodes [1] [2]. In practice, Livepeer acts as a permissionless GPU compute marketplace for media – delivering live streaming, transcoding, and now generative AI video affordably. For example, a 2025 report calls Livepeer a "global compute network for AI video" that enables streaming and real-time video inference at lower cost [2] [1].

Livepeer's recent **Daydream** platform extends the network to real-time AI video. Daydream lets creators turn live camera input into stylized animations with a simple text prompt [3] [4]. In other words, it can instantly transform a webcam feed into a fully-animated world [5]. Artists are already embracing this – for example, DJ/artist MetaDJ uses Daydream to invite fans into his live sets, "blurring the line between reality and digital artistry" [6] [3].

The native token **LPT** secures and governs the network. Holding LPT lets users stake and run nodes, earn fees, and vote on upgrades [7] [8]. In practice, *Orchestrators* (node operators) bond LPT to process jobs, while *Delegators* stake LPT behind trusted operators to bolster security and decentralization [8] [7]. Both roles earn rewards: token emissions and a share of ETH fees generated by work. Staking LPT thus ties you directly to the network's success.

<Tip>Staking LPT makes you a network supporter: you earn rewards while helping secure and decentralize the protocol [7] [8].</Tip>

Explore More

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Protocol Overview

The Livepeer protocol is the software layer that coordinates decentralized video and AI processing. Its core is implemented as Ethereum smart contracts (now on Arbitrum L2 after the Confluence upgrade) complemented by off-chain nodes and services [9]. The protocol enforces staking, bonding, slashing, and reward rules on-chain, while specialized nodes handle the actual video jobs. Official docs note:

"Livepeer is a decentralized infrastructure protocol… nodes including Gateways (formerly broadcasters), Orchestrators, and Transcoders" [9] [10] .

- **Gateways (Broadcasters):** These demand-side nodes submit video/AI jobs to Livepeer. A gateway takes a video stream (or AI task), creates on-chain payment tickets, and routes the work to an Orchestrator. It then collects the transcoded or AI-processed result. Gateways are typically stateless routers that pay for compute.
- **Orchestrators:** These supply-side nodes bond LPT and accept jobs from gateways. An orchestrator advertises which video/audio renditions it can provide, and claims work up to its bonded stake. When it receives a job, it divides the work among its workers (transcoders or AI workers). The orchestrator must also periodically call the on-chain reward function to distribute newly minted LPT to itself and its delegators [11] .
- **Workers (Transcoders):** These are the actual compute units (often GPUs) that perform video encoding or AI inference. In legacy terms, they were called Transcoders. Today they execute the heavy lifting, and they are typically managed by an Orchestrator node. Once a worker finishes processing a segment, the orchestrator returns the result to the Gateway.
- **Delegators:** LPT holders who do not run nodes. Delegators stake their tokens on existing Orchestrators to secure the network. Their stake increases an orchestrator's bonded weight, which can route more work to it. In return, delegators passively earn a share of the orchestrator's rewards (both LPT emissions and ETH fees) proportional to their stake [12] [13] .

The **job routing flow** can be summarized as: a Gateway creates a job request → an Orchestrator accepts the job → the Orchestrator assigns it to a Worker node → the Worker processes the segment → the result is returned to the Gateway. Delegators sit "behind" the orchestrator by staking LPT (shown below). The orchestrator earns ETH fees from the Gateway, and can share a portion of those fees with delegators based on its fee-sharing policy [14] [15] . Smart contracts on Arbitrum enforce each step, ensuring trustless cooperation.

```
flowchart LR
  B[Gateway (Broadcaster)] --> O[Orchestrator]
  O --> W[Worker (Transcoder)]
  W --> B
  D[Delegator] -.-> O
```

This diagram shows a simplified Livepeer job: the Gateway (left) sends work to an Orchestrator (middle), which directs a Worker (right) to process it, and then returns results back to the Gateway. Delegators (dotted line) back the Orchestrator with staked LPT [10] [15] .

Actor Overview

Livepeer involves multiple actors, both on-chain (protocol) and off-chain (community). Here's a breakdown of key roles:

- **Orchestrator:** A bonded node operator that claims work on-chain. Orchestrators maintain active LPT stake, set fee-sharing percentages, and call the on-chain reward function each round. They earn ETH fees from broadcasters and LPT from inflation. (Messari: *"Orchestrators (node operators) route transcoding & AI jobs… earning ETH fees and LPT rewards"* [15] .)

- **Transcoder / Worker:** A compute node (often a GPU) that performs the actual encoding or inference. Workers do not hold stake themselves; they operate under an Orchestrator.
- **Gateway (Broadcaster):** A stateless demand-side node. Gateways submit video or AI jobs to the network and pay the node that completes the work. They effectively bridge an input stream into Livepeer, handling micropayment tickets on-chain [10].
- **Delegator:** A token holder who stakes LPT behind one or more orchestrators. Delegators strengthen network security and decentralization. In exchange, they receive a portion of the node's rewards (both mint emissions and usage fees) according to their stake. As one Livepeer blog notes, *"Delegators earn a share of both token emissions (while they last) and usage-based fees"* [13] [16].

- **Livepeer Foundation:** A non-profit entity formed to guide ecosystem growth. The Foundation coordinates long-term strategy and funding, runs community programs (like advisory boards), and helps align core protocol development with the open-source community [17].

- **AI Artists & Content Creators:** Performers and creators who use Livepeer's tools (like Daydream) to produce content. For example, MetaDJ used Daydream to let fans shape the live visuals in real time [6]. Any streamer, musician, or videographer can leverage the network for novel AI-enhanced experiences.

- **Developers & Builders:** Software teams building on Livepeer. This includes decentralized video apps (e.g. on Web3 social networks) and infrastructure tools. Messari notes Livepeer is used by projects ranging from decentralized streaming platforms to tokenized music apps [18].
- **Gateway Operators:** Organizations running public Livepeer gateway services. For example, *Livepeer Cloud* (a community-run gateway) and *Livepeer Studio* (a production-grade API gateway) provide easy on-ramps to the network [19]. They ensure reliable, scalable access for creators and apps.
- **Special Purpose Entities (SPEs):** Community-approved teams funded by the on-chain treasury. SPEs tackle public goods for Livepeer (e.g. building dev tools, infrastructure). For example, Livepeer's ecosystem has active SPEs like *LiveInfra* and *LISAR* working on network upgrades and on-ramp infrastructure [20].

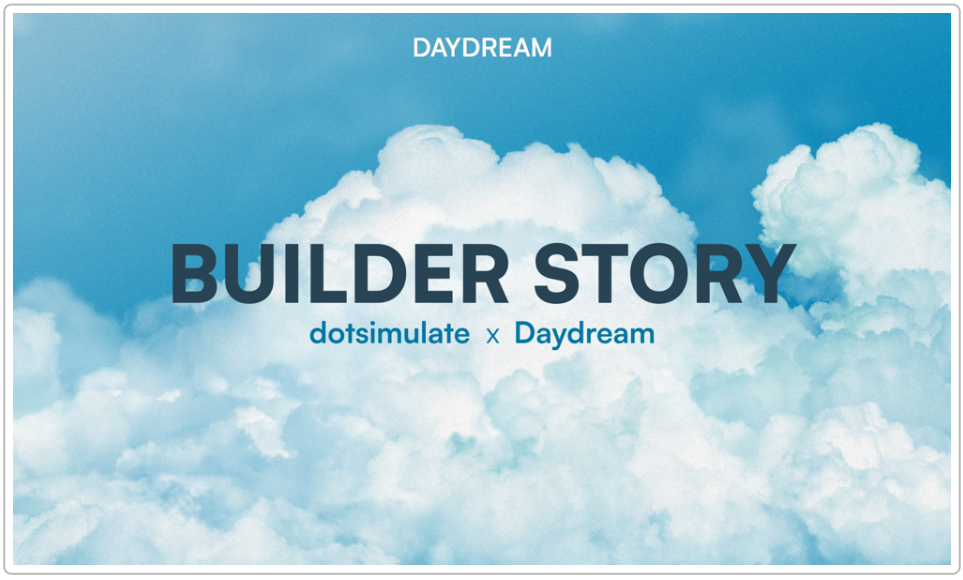

Figure: An example of a Livepeer "builder" in action. The *StreamDiffusionTD* operator (shown above) is a TouchDesigner plug-in created by a community developer. It streams live inputs into a generative AI

model via Daydream's API, offloading heavy GPU inference to the cloud [21] [22]. This allows artists to run professional-quality real-time video effects without a top-end PC.

Livepeer Governance

Livepeer governance combines on-chain voting with community processes. Protocol changes are made via **Livepeer Improvement Proposals (LIPs)**. A typical flow is:

- The community discusses and drafts a LIP (often on the Livepeer forum).
- Once a LIP is finalized, anyone with ≥100 LPT can submit it on-chain to the Governor contract [23].
- LPT holders vote in a stake-weighted on-chain referendum. If the vote passes, the change is executed automatically by the smart contract framework [23].

Recent LIPs illustrate this process. In late 2023, the community passed a bundle creating the Livepeer treasury. LIP-89 deployed a custom OpenZeppelin Governor (with a 100 LPT proposal threshold and stake-weighted voting) [23]. LIP-90 established that the treasury should fund public goods (primarily via Special Purpose Entities) [24]. Subsequent LIP-92 set the on-chain revenue allocation (sending 10% of new LPT emissions into the treasury) [23] [25]. Today, Livepeer maintains an on-chain treasury that accrues 10% of all LPT block rewards, with any LPT holder able to propose how to spend it (subject to voting) [25] [23].

Governance Highlights: - **LIPs & Voting:** Protocol upgrades and parameter changes are formalized as LIPs. After community vetting, an on-chain vote is held. Stake-weighted voting ensures larger delegations have proportional say [23]. - **On-Chain Treasury:** A portion of token emissions flows to a community treasury (created by LIPs). This treasury is meant to fund ecosystem-wide projects (public goods). Livepeer's social consensus is that treasury funds should primarily go to SPEs, which then deploy them to specific initiatives [24] [25]. - **Livepeer Foundation:** The Livepeer Foundation (formed 2024) is a neutral entity that fosters long-term strategy and ecosystem coordination. It establishes advisory boards, publishes roadmaps, and helps shepherd core development and on-chain proposals [17]. Importantly, the Foundation facilitates community participation but **does not override token-holder authority** – only on-chain votes by LPT holders can enact funding or protocol changes.

```
flowchart TD
  A[Community Discussion] --> B[LIP Proposal (Forum/GitHub)]
  B --> C[On-Chain Vote (100 LPT threshold)]
  C --> D{Vote Outcome}
  D -- Approved --> E[Execute Change (Protocol/Treasury)]
  D -- Rejected --> F[Revise or Abandon]
```

This flowchart shows the governance process: community brainstorming leads to a LIP submission, which triggers an on-chain vote. A successful vote executes the change, while a rejected vote sends the proposal back for revision.

Livepeer Token Economics

Livepeer's economics balance inflationary rewards with real usage fees.

- **Inflation & Staking:** Livepeer uses dynamic inflation to incentivize staking. If under 50% of tokens are staked, inflation increases; if above 50%, it decreases [26]. Currently roughly half of all LPT is staked, yielding an inflation of about ~25–30% annual (~30k LPT/day) [27]. Over time, as more tokens lock up in staking, inflation will gradually taper off, shifting more reward emphasis to network usage fees [28] [16].
- **Usage-Based Fees:** Node operators earn ETH fees from each video or AI job processed. Gateways (broadcasters) pay these fees segment-by-segment. Orchestrators set a fee-sharing percentage to split collected ETH with their delegators. As network usage has grown, delegators increasingly benefit from these real fees. As Livepeer's X account notes, "real usage is generating real fees, and delegation links you directly to that value" [16] [28].
- **Reward Distribution:** Each staking round, new LPT tokens are distributed by the `reward()` function. 90% of inflation goes to a reward pool for active nodes and their delegators, proportional to their bonded stake [11]. For example, if an orchestrator has 100 LPT (70 self + 30 delegated), and total stake is 500 across all nodes, it earns 20% of that round's rewards (14 LPT to operator, 6 LPT to delegators, if it set fee cuts). 10% of LPT inflation is diverted to the community treasury [25]. User fees (ETH) likewise flow to active nodes – split according to each orchestrator's fee share setting.

```
flowchart LR
  LPT[LPT Emissions] --> RewardPool[Reward Pool (90%)]
  RewardPool --> OR(Orchestrators)
  RewardPool --> DLG(Delegators)
  LPT -->|10%| TRE(Treasury)
  ETH[Usage Fees (ETH)] --> OR
  ETH --> DLG
```

The above diagram illustrates token flows: new LPT (emissions) largely goes into a reward pool (split between orchestrators and delegators by stake). A fraction is allocated to the treasury. Usage fees (ETH) collected from jobs also flow to nodes and their delegators. Delegators, therefore, earn from both inflationary rewards and rising fee revenue as the network grows.

<Check>Delegators earn both inflationary LPT rewards and a share of user fees, aligning their stake with network demand [13] [16].</Check>

Livepeer Whitepaper

Livepeer's original 2017 whitepaper envisioned a decentralized video transcoding service. It described how tokens incentivize nodes to process video cheaper than cloud providers. In practice, Livepeer has delivered on this: it has enabled up to ~10× cost savings over centralized solutions [29]. Early on, hours of video were transcoded weekly, proving the concept of a crowd-sourced GPU network.

Since then, Livepeer has evolved with Web3 trends. Major upgrades like *Confluence* (2023) moved the protocol to Arbitrum for scalability. In late 2024, Livepeer introduced its **Cascade** vision to focus on real-

time AI video pipelines [30]. In other words, the network is now explicitly targeting generative AI workloads in addition to transcoding. This is reflected by projects like Daydream and artist tools.

Livepeer is also developing an *AI Video Subnet* (alpha) for decentralized AI inference [31]. The AI Subnet adds new node types (AI Gateways/Orchestrators) and integrates text-to-image/video models into the protocol. For example, users can send live video through a live AI model to generate stylized output, all coordinated by Livepeer's network. These innovations extend the original whitepaper's idea – using staked tokens to coordinate compute – into the AI era.

```
flowchart TB
    2017["2017: Whitepaper – Decentralized video transcoding"]
    2023["2023: Confluence (Arbitrum) & Delta upgrades"]
    2024["2024: Launch Daydream & Cascade AI vision"]
    2025["2025: AI Video Subnet (alpha) for generative pipelines"]
    2017 --> 2023 --> 2024 --> 2025
```

<Info>The Livepeer whitepaper laid out a decentralized streaming network; today's protocol builds on that foundation with AI capabilities. Staked nodes still coordinate workloads, but now those workloads increasingly include AI video inference (as seen in Daydream and StreamDiffusion integrations) [29] [31].</Info>

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